Flat Rotation Curves from Quantum Relative Entropy: Running Newton's Constant as Information Processing

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[**TO WRITE:** Final abstract after calculations are complete.]

We show that the quantum relative entropy (QRE) framework developed in Papers 1 and 2—where entropy production $\Sigma = D(\rho\|\sigma) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma))$ quantifies the arrow of time—naturally explains flat galaxy rotation curves without dark matter particles. The key insight combines two established results: (i) Dorau and Much's derivation of semiclassical Einstein equations from QRE at a fixed scale (PRL 2025), and (ii) Casini and Huerta's proof that renormalization group (RG) flow is equivalent to data processing inequality (DPI) monotonicity of QRE. Together, these imply that the gravitational coupling G inherits a scale dependence $G(k)$ from the RG flow of the gravitational quantum channel. With the marginal anomalous dimension $\eta = 1$ (Kumar 2025), this yields a logarithmic correction $\Phi(r) = -G_N M/r + (2G_N M k_*/\pi)\ln(r/r_0)$ whose universal coefficient $2/\pi$ produces flat rotation curves at $r \gg r_c \approx 37$ kpc. We reinterpret the SPARC galaxy fits of Gubitosi et al. (2024) within the Σ framework, where the running parameter $\bar{\nu}$ is identified with the QRE drop per RG step, transforming it from a free parameter into an information-theoretic quantity. Predictions for BIG-SPARC ($\sim$ 4000 galaxies) and honest limitations (CMB, Bullet Cluster) are discussed.

INTRODUCTION

Content outline

[KNOWN] *Papers 1–2 recap.*—Paper 1 [1] established the temporal asymmetry parameter

$$\tau = 1 - F\big(\rho,\, \tilde{\mathcal{R}}_{\sigma,\mathcal{N}}(\mathcal{N}(\rho))\big), \tag{1}$$

quantifying the arrow of time as Petz recovery failure, with the universal bound $\tau \leq 1 - \exp(-\Sigma/2)$. Paper 2 [2] showed that in the strong gravitational field of a static, spherically symmetric mass, the entropy production takes the form $\Sigma_{\text{grav}} = r_s/r$, connecting the Petz recovery fidelity to the gravitational redshift: $F_{\text{grav}} = \sqrt{g_{00}} = \exp(-r_s/(2r))$.

[KNOWN] *The dark matter puzzle.*—Galaxy rotation curves flatten at large radii: $v(r) \to V_0 = \text{const}$ for $r \gtrsim 10$ kpc [8, 9], implying $M_{\text{dyn}}(r) \propto r$, far exceeding the visible baryonic mass M_{bar}. The standard explanation posits invisible dark matter (DM) particles [10], but despite four decades of searches, no DM particle has been detected directly [11, 12].

[NEW SYNTHESIS] *Our proposal.*—We argue that flat rotation curves emerge naturally from the *same* $\Sigma = D(\rho\|\sigma) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma))$ framework. The gravitational quantum channel $\mathcal{N}_{\text{grav}}$ is not scale-independent: quantum corrections make $\mathcal{N}_{\text{grav}}(k)$ "noisier" at low momenta k (large distances), causing G to run. This is not a new term added by hand, but the natural consequence of evaluating the *same* Σ formula with the full quantum-corrected gravitational channel.

[**TO WRITE:** Paragraph on analogy to $F = ma$ with different force laws giving different orbits, springs, fluids. One equation, different boundary conditions.]

[**TO WRITE:** Paragraph on paper structure: Sec. II reviews Dorau-Much; Sec. III connects RG to DPI; Sec. IV derives running G and logarithmic correction; Sec. V compares with SPARC data; Sec. VI gives predictions and limitations; Sec. VII discusses implications.]

FROM QUANTUM RELATIVE ENTROPY TO EINSTEIN EQUATIONS

Content outline

[KNOWN] *Araki–Uhlmann modular theory.*—For a von Neumann algebra $\mathcal{M}$ with two faithful normal states ω_0 (vacuum) and ω_ϕ (coherent excitation), the Araki relative entropy is [17]

$$S(\omega_\phi\|\omega_0) = -\langle \Omega_\phi, \ln \Delta_{\omega_\phi|\omega_0}\, \Omega_\phi\rangle, \tag{2}$$

where $\Delta_{\omega_\phi|\omega_0}$ is the relative modular operator. This is the algebraic QFT generalization of the finite-dimensional quantum relative entropy $D(\rho\|\sigma) = \text{Tr}[\rho(\ln\rho - \ln\sigma)]$.

[KNOWN] *Dorau–Much derivation (PRL 2025).*—The derivation proceeds in five steps [4]:

1. Start with a local Rindler horizon (equivalence principle): any point in spacetime admits an approximate Rindler wedge.

2. Compute $S(\omega_\phi\|\omega_0)$ between the vacuum ω_0 and a coherent excitation ω_ϕ of a scalar field on the right Rindler wedge:

$$S(\omega_\phi\|\omega_0) = -2\pi \int_{\mathcal{H}_B^R} U\,(\partial_U \phi)^2\, dU\, d\text{vol}_S. \tag{3}$$

3. Identify the area variation via the Bekenstein–Hawking relation: $\delta A = 4G\, S(\omega_\phi \| \omega_0)$.

4. Apply the Raychaudhuri equation to relate δA to the Ricci tensor.

5. *Result*: the semiclassical Einstein equations follow:

$$\boxed{G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle.} \tag{4}$$

[**TO WRITE:** Paragraph on significance: This establishes that QRE is the *microscopic origin* of Einstein's equations, extending Jacobson's 1995 thermodynamic derivation [18] to the quantum information setting. Crucially, this derivation holds at a *fixed scale*—it does not account for quantum corrections that appear when the theory is probed at different energy/distance scales.]

The fixed-scale limitation.—Eq. (4) uses the *bare* Newton constant G. It does not incorporate the scale dependence of G that arises from quantum loop corrections to the gravitational channel. Our central claim is that this scale dependence, when included via the data processing inequality, naturally produces running $G(k)$ and flat rotation curves.

RENORMALIZATION GROUP FLOW AS DATA PROCESSING

Content outline

[KNOWN] *Data processing inequality (DPI).*—For any quantum channel $\mathcal{N}$ and states ρ, σ [15, 16],

$$D\big(\mathcal{N}(\rho) \big\| \mathcal{N}(\sigma)\big) \leq D(\rho \| \sigma). \tag{5}$$

Equality holds if and only if the Petz recovery map $\mathcal{R}_{\sigma,\mathcal{N}}$ exactly recovers ρ [13, 14].

[KNOWN] *RG flow as a quantum channel.*—Casini and Huerta [5] proved that renormalization group (RG) flow from UV to IR can be formulated as a quantum channel: the operation of integrating out high-energy modes is a CPTP map $\mathcal{N}_{\mathrm{RG}} : \rho_{\mathrm{UV}} \mapsto \rho_{\mathrm{IR}}$. By the DPI,

$$D(\rho_{\mathrm{IR}} \| \sigma_{\mathrm{IR}}) \leq D(\rho_{\mathrm{UV}} \| \sigma_{\mathrm{UV}}). \tag{6}$$

This monotonicity of QRE under RG flow provides:

- An alternative proof of the c-theorem in $d = 2$ (Zamolodchikov);
- In $d > 2$: the coefficient of the area-law term in entanglement entropy *decreases* along RG flow;
- A unified "entropic" proof of irreversibility theorems (a-theorem in $d = 4$, F-theorem in $d = 3$).

[NEW SYNTHESIS] *Scale-dependent* Σ.—Define the entropy production at scale k:

$$\Sigma(k) = D(\rho \| \sigma) - D\big(\mathcal{N}_{\mathrm{grav}}(k)(\rho) \big\| \mathcal{N}_{\mathrm{grav}}(k)(\sigma)\big). \tag{7}$$

Here $\mathcal{N}_{\mathrm{grav}}(k)$ is the gravitational channel evaluated at RG scale k: it describes the information processing of gravity *as seen at resolution* $1/k$.

[NEW SYNTHESIS] *Key observation: as k decreases (probing larger distances), $\mathcal{N}_{\mathrm{grav}}(k)$ becomes a "noisier" channel.* This is because integrating out more UV modes increases the effective dissipation. By the DPI, $\Sigma(k)$ can only increase or stay the same as k decreases:

$$\Sigma(k_1) \leq \Sigma(k_2) \quad \text{for } k_1 > k_2. \tag{8}$$

More entropy production at larger scales = more information loss = stronger effective gravity. This is the information-theoretic origin of running G.

[**TO WRITE:** Paragraph connecting to Casini-Huerta's c-theorem: the decrease of QRE along RG flow is the *same* mathematical structure as the DPI for the gravitational channel. RG irreversibility *is* data processing.]

[NEW SYNTHESIS] *Observer dependence.*—Basso, Maziero, and Celeri [36] proved that entropy production in curved spacetime is observer-dependent, validating the coordinate dependence of $\Sigma(k)$. De Vuyst et al. [38] confirmed this via quantum reference frames. This is not a weakness of the framework but a *physical feature*: different observers probe gravity at different scales and see different effective channels $\mathcal{N}_{\mathrm{grav}}(k)$.

Galley, Giacomini, and Selby [37] proved that *any* consistent coupling between classical gravity and quantum matter is fundamentally irreversible ($\Sigma > 0$). This provides model-independent justification for the premise that gravitational channels produce entropy, independent of the specific form of $\mathcal{N}_{\mathrm{grav}}$.

RUNNING G AND THE LOGARITHMIC CORRECTION

Content outline

[KNOWN] *IR running of Newton's constant.*—In the effective field theory of gravity, quantum loop corrections induce a scale-dependent Newton coupling [19, 20]:

$$G(k) \sim G_N \left(\frac{k_*}{k}\right)^{\eta} \quad \text{for } k \ll k_*, \tag{9}$$

where η is the anomalous dimension governing the IR flow and k_* is a crossover momentum scale.

[KNOWN] *Why $\eta = 1$ is special (marginal).*—Kumar [6] showed that the Fourier transform of $k^{-(2+\eta)}$ in three spatial dimensions gives:

- $\eta \neq 1$: power-law $\sim r^{\eta-1}$ (IR-irrelevant for $\eta < 1$, spoils scale invariance for $\eta > 1$);
- $\eta = 1$: logarithmic $\sim \ln(r)$ — the *unique* marginal, scale-invariant deformation consistent with rotational symmetry and locality.

[**ASSUMED:** $\eta = 1$ from marginality/dimensional analysis. Not yet derived from QRE channel properties. This is the main theoretical gap. See Sec. for discussion.]

[KNOWN] *Modified Newtonian potential.*—With $G(k) = G_N[1 + (k_*/k)]$ for $\eta = 1$, the gravitational potential in position space is [6]

$$\boxed{\Phi(r) = -\frac{G_N M}{r} + \frac{2G_N M k_*}{\pi} \ln\left(\frac{r}{r_0}\right),} \tag{10}$$

with r_0 an integration constant. The force law becomes

$$F(r) = -\frac{G_N M}{r^2} - \frac{2G_N M k_*}{\pi} \frac{1}{r}. \tag{11}$$

At $r \gg r_c = 1/k_*$, the $1/r$ force dominates over the Newtonian $1/r^2$.

[KNOWN] *Flat rotation curves.*—The circular velocity is

$$v^2(r) = \frac{G_N M_{\rm bar}}{r} + \frac{2G_N M_{\rm bar} k_*}{\pi}, \tag{12}$$

which asymptotes to $v^2 \to 2G_N M_{\rm bar} k_*/\pi$ at $r \gg r_c$. The asymptotic velocity V_0 determines k_* via [6]

$$k_* = \frac{\pi V_0^2}{2G_N M_{\rm bar}}. \tag{13}$$

[KNOWN] *Universality of crossover scale.*—For three SPARC galaxies, Kumar [6] found:

Galaxy	$M_{\rm bar}$ [$M_\odot$]	V_0 [km/s]	r_c [kpc]
S:700624	1.01×10^{12}	271.4 ± 2.6	37.5 ± 0.7
S:700916	5.85×10^{11}	210.4 ± 2.4	36.2 ± 0.8
S:705253	3.55×10^{11}	158.9 ± 1.9	38.5 ± 0.9

The remarkably consistent $r_c \approx 37$ kpc across galaxies spanning a factor of 3 in mass suggests a universal IR scale.

[KNOWN] *Free parameters.*—Only 2 parameters at the galactic scale: k_* (crossover scale) and G_N (Newton's constant). The coefficient $2/\pi$ is universal and regulator-independent.

[NEW SYNTHESIS] *Information-theoretic interpretation.*—In the Σ language, Eq. (10) translates to

$$\Sigma_{\rm grav}(r) = \frac{r_s}{r} + \frac{4k_* r_s}{\pi} \ln\left(\frac{r}{r_0}\right), \tag{14}$$

where $r_s = 2G_N M/c^2$. The logarithmic term represents *additional information loss* due to quantum corrections to the gravitational channel at large distances. The channel $\mathcal{N}_{\rm grav}(k)$ at low k (large r) loses more information per unit QRE, producing extra Σ, which manifests as extra gravitational potential.

[**TO DERIVE:** Show explicitly that $\Sigma(k)$ computed from the Dorau–Much framework at each RG scale k reproduces Eq. (14). This requires defining $\mathcal{N}_{\rm grav}(k)$ using the RG-improved Hamiltonian $H(k)$ with $G(k)$ and computing the QRE drop. This is the central calculation of the paper.]

[**TO DERIVE:** Show that the coefficient $2/\pi$ emerges from the channel properties of $\mathcal{N}_{\rm grav}$ at the marginal point $\eta = 1$. This would promote the coefficient from "universal in Kumar's EFT sense" to "universal in the information-theoretic sense."]

COMPARISON WITH SPARC DATA

Content outline

[KNOWN] *The SPARC database.*—The Spitzer Photometry and Accurate Rotation Curves (SPARC) catalog [21] contains 175 nearby disk galaxies with 3.6 μm photometry and high-quality rotation curves, providing the standard benchmark for testing gravity theories at galactic scales.

[KNOWN] *RGGR fits.*—Gubitosi, Piattella, and Casarini [7] fitted 100 SPARC galaxies using the renormalization group improved gravity (RGGR) model with running coupling

$$G(\mu) = \frac{G_0}{1 + \nu \ln(\mu^2/\mu_0^2)}, \tag{15}$$

giving effective circular velocity

$$v^2_{\rm RGGR}(r) = v_N^2(r)\left[1 - \frac{c^2 \bar{\nu}}{\Phi_N(r)}\right], \tag{16}$$

where $\bar{\nu} = \nu\alpha$ is a phenomenological parameter. Using BIC analysis: 47 galaxies prefer RGGR over NFW, 40 prefer NFW, 13 inconclusive.

[KNOWN] *Mass-dependent $\bar{\nu}$.*—The RGGR parameter $\bar{\nu}$ correlates near-linearly with galactic baryonic mass:

Galaxy type	$\bar{\nu}$
Early type	$\sim 10^{-6}$
Spiral	$\sim 10^{-7}$
Late/Starburst	$\sim 10^{-8}$
Solar system	$\sim 10^{-17}$

[NEW SYNTHESIS] *Reinterpretation within Σ framework.*—The RGGR parameter $\bar{\nu}$ is identified with the QRE drop per RG step of the gravitational channel:

$$\bar{\nu} = \left.\frac{\Delta\Sigma}{\Delta \ln k^2}\right|_{k \sim k_*}, \tag{17}$$

transforming $\bar{\nu}$ from a free parameter into an information-theoretic observable. The mass dependence of $\bar{\nu}$ then follows from the mass dependence of $\Sigma_{\rm grav}$: more massive galaxies have larger r_s, hence larger QRE drop per RG step.

[**TO DERIVE:** Compute $\Delta\Sigma/\Delta \ln k^2$ from the gravitational channel and verify that the mass scaling matches the observed $\bar{\nu}(M_{\rm bar})$ correlation.]

[KNOWN] *The Radial Acceleration Relation (RAR).*—The tightest empirical constraint is the RAR [22]:

$$g_{\rm obs} = \frac{g_{\rm bar}}{1 - \exp\left(-\sqrt{g_{\rm bar}/a_0}\right)}, \tag{18}$$

with $a_0 = 1.20 \times 10^{-10}$ m/s^2, observed scatter 0.13 dex over 2693 data points in 153 galaxies.

[NEW SYNTHESIS] *RAR from running* Σ.—The logarithmic correction in Eq. (10) produces an additional acceleration $g_{\rm extra} = 2G_N M_{\rm bar} k_*/(\pi r)$, which becomes comparable to $g_{\rm bar} = G_N M_{\rm bar}/r^2$ precisely when $g_{\rm bar} \lesssim 2k_* c^2/\pi$. Identifying

$$a_0 \sim \frac{2k_* c^2}{\pi}, \tag{19}$$

we recover the correct order of magnitude for the MOND acceleration scale. With $k_* \approx 2.7 \times 10^{-2}$ kpc^{-1}, this gives $a_0 \sim 1.7 \times 10^{-10}$ m/s^2, within a factor of 1.4 of the observed value.

[**TO WRITE:** Paragraph on the coincidence $a_0 \sim cH_0$ (Milgrom [25]) and its possible information-theoretic origin: the de Sitter horizon entropy sets the IR cutoff for the gravitational channel, connecting the acceleration scale to cosmology.]

PREDICTIONS, TESTS, AND LIMITATIONS

Content outline

[NEW SYNTHESIS] *Prediction 1: BIG-SPARC.*—The upcoming BIG-SPARC catalog [23] with $\sim$ 4000 galaxies (20$\times$ larger than SPARC) provides a stringent test. Our prediction: the crossover scale $r_c \approx 37$ kpc should remain universal across the expanded sample, and the RGGR parameter $\bar{\nu}$ should maintain its near-linear correlation with baryonic mass.

[NEW SYNTHESIS] *Prediction 2: Dwarf galaxies.*—Dwarf spheroidal galaxies are the strongest test case, as they have the largest mass discrepancies. Ghari and Haghi [24] found that Verlinde's emergent gravity (which shares the same information-theoretic structure as our approach) is favored over MOND at 5.2σ for 23 dwarf spheroidals. Our prediction: the Σ framework should reproduce these fits with the *same* universal parameters.

[NEW SYNTHESIS] *Prediction 3: Solar system screening.*—At solar system scales ($r \sim 1$ AU $\ll r_c \approx$ 37 kpc), the logarithmic correction in Eq. (10) contributes $\delta\Phi/\Phi_N \sim 2k_* r/\pi \sim 10^{-15}$, far below current PPN constraints. The theory is automatically screened at small scales.

[KNOWN] *Limitation 1: CMB acoustic peaks.*—Running G alone *cannot* reproduce the CMB acoustic peak structure without dark matter [6]. The IR running effect is negligible at $z \sim 1100$ when the universe was $\sim$ 380,000 years old. This is a severe limitation.

[**TO WRITE:** Honest discussion: Does this rule out the approach? Not necessarily—the CMB constraint may require *additional* physics beyond running G (e.g., sterile neutrinos, primordial fluctuations, or modifications to the gravitational channel at earlier epochs). The Σ framework is open to incorporating such effects if they arise naturally from the channel properties. But we cannot claim to replace dark matter entirely.]

[KNOWN] *Limitation 2: Bullet Cluster.*—The Bullet Cluster (1E 0657-56) shows spatial separation between the gravitational potential (traced by lensing) and the baryonic gas (traced by X-rays) [26]. No systematic analysis of the Σ framework at cluster scales has been performed. This is a well-known challenge for all modified gravity theories.

[KNOWN] *Limitation 3: Structure formation.*—Cosmological structure formation at high redshift requires matter perturbations that grow faster than baryons alone can provide. Running G may help [20], but a full perturbation theory within the Σ framework has not been developed.

[**TO WRITE:** Summary table of what the approach explains, what it does not, and what remains open:]

TABLE I. Status of observational tests.

Observable	**Status**	**Notes**
Flat rotation curves	Yes	3 galaxies (Kumar)
SPARC (100 galaxies)	Yes	RGGR fits (Gubitosi+)
RAR (a_0 scale)	Partial	Order-of-magnitude
Solar system PPN	Yes	Auto-screened
Dwarf spheroidals	Promising	Verlinde 5.2σ
Weak lensing (KiDS)	Promising	Brouwer+ 2021
CMB acoustic peaks	**No**	Requires extra physics
Bullet Cluster	Unknown	Not analyzed
Structure formation	Unknown	No perturbation theory
Lyman-α forest	Unknown	Not analyzed

KEY THEORETICAL GAPS

Content outline

[**TO WRITE:** Honest assessment of the three main gaps.]

Gap 1: Deriving $\eta = 1$ from QRE.—The anomalous dimension $\eta = 1$ is currently argued from dimensional analysis and marginality [6], not derived from the channel properties of $\mathcal{N}_{\text{grav}}$. A first-principles derivation from QRE would require showing that $\eta = 1$ is the unique value at which the DPI bound for the gravitational channel is "marginally" saturated—i.e., the boundary between recoverable and non-recoverable regimes of the channel.

[SPECULATIVE] *Possible argument*: In the information-theoretic framework, $\eta < 1$ corresponds to a channel where the DPI deficit *decreases* with scale (the channel becomes "less noisy" in the IR), which would mean gravity *weakens* at large distances—unphysical. $\eta > 1$ corresponds to *super-logarithmic* growth of Σ, spoiling the perturbative expansion. $\eta = 1$ is the unique marginal case where Σ grows logarithmically, maintaining the delicate balance between gravity strengthening (as required by observations) and perturbative control.

[SPECULATIVE] *New conjecture*: Applying the Fawzi–Renner recovery bound to the gravitational RG flow suggests

$$\eta \leq 1 - \exp(-\Sigma_{\text{RG}}/2), \tag{20}$$

where Σ_{RG} is the entropy production per RG step. For large Σ_{RG}, this gives $\eta \leq 1$, making $\eta = 1$ the maximal allowed value. In flat space ($\Sigma_{\text{RG}} = 0$), $\eta = 0$, which is the correct UV fixed point.

Gap 2: Defining $\mathcal{N}_{\text{grav}}(k)$ rigorously.—The gravitational channel at scale k can be defined via:

- Option A: RG-improved Hamiltonian $H(k)$ with $G(k)$ replacing G_N in the Pikovski-type coupling;
- Option B: One-loop corrected graviton propagator in the effective field theory;
- Option C: Modular channel with scale-dependent modular Hamiltonian (most rigorous but hardest to compute).

A rigorous definition is needed to compute $\Sigma(k)$ explicitly.

Gap 3: The acceleration scale $a_0 \sim cH_0$.—The coincidence between the MOND acceleration scale $a_0 \approx 1.2 \times 10^{-10}$ m/s^2 and $cH_0 \approx 6.9 \times 10^{-10}$ m/s^2 is suggestive but unexplained within the Σ framework. Possible origin: the de Sitter horizon entropy $S_{\text{dS}} = \pi c^3/(G\hbar H_0^2)$ sets the IR cutoff for the gravitational channel, making $k_{\min} \sim H_0/c$ and $a_0 \sim c^2 k_{\min}$. This connects galactic dynamics to cosmology through the *same* QRE structure.

DISCUSSION

Content outline

[NEW SYNTHESIS] *What is genuinely new.*—Three things:

1. *The connection*: QRE at each scale $\to$ running G $\to$ rotation curves. Dorau-Much [4] gives Einstein equations from QRE at a fixed scale. Casini-Huerta [5] shows QRE monotonicity = RG irreversibility. We combine them: QRE at each scale gives scale-dependent Einstein equations with running G.
2. *The reinterpretation*: The RGGR parameter $\bar{\nu}$ is not a free parameter but the QRE drop per RG step, connecting phenomenological galaxy fitting to fundamental information theory.
3. *The unification*: Papers 1, 2, and 3 use the *same* Σ framework at different scales. $\Sigma = 0$ (Paper 1, QI), $\Sigma = r_s/r$ (Paper 2, strong field), $\Sigma = r_s/r +$ log correction (Paper 3, weak field).

What is NOT new.—Running G has been proposed since Reuter and Weyer (2004) [20]. Kumar [6] derived the logarithmic correction. Gubitosi et al. [7] fitted SPARC galaxies. Our contribution is the *information-theoretic grounding*: why G runs, what the running means in terms of quantum channels, and how it connects to the broader Σ framework.

[**TO WRITE:** Paragraph on what this does NOT achieve: we do not explain CMB, Bullet Cluster, or structure formation. We are honest that "replacing dark matter" is too strong a claim. What we show is that *at galactic scales*, running Σ from QRE provides a viable alternative that is competitive with NFW halos and has a deeper theoretical motivation.]

Comparison with other information-theoretic approaches.— Several groups have connected information theory to dark matter:

- Verlinde [27]: Volume-law entanglement entropy in de Sitter $\to$ apparent dark matter. Complementary to our approach; may be the same physics in different language.
- Bianconi [28]: Gravity action = QRE, metric as density matrix. However, the GfE vacuum solutions give Schwarzschild + $O(r^{-4})$ corrections [39], not the logarithmic modification needed at galactic scales. Our approach via running $\Sigma(k)$ does not depend on GfE.
- Smolin [29]: MOND from quantum gravity regime below de Sitter temperature.

[NEW SYNTHESIS] *Connection to Papers 1–2 and preview of Paper 4.*— The logical chain is now:

Paper 1: $\Sigma = 0 \Leftrightarrow$ perfect recovery,

Paper 2: $\Sigma_{\rm grav} = r_s/r \;\rightarrow\;$ exponential metric,

Paper 3: $\Sigma_{\rm grav}(k) = r_s(k)/r \;\rightarrow\;$ flat curves,

Paper 4: $\Sigma = D(\rho_{\rm st}\|\rho_{\rm m}) \;\rightarrow\;$ all three as limits.

Paper 4 [3] will argue that these are not three separate theories but three solutions of *one equation* $\Sigma = D(\rho_{\rm spacetime}\|\rho_{\rm matter})$ under different boundary conditions, analogous to $F = ma$ giving planetary orbits, spring oscillations, and fluid dynamics.

The author thanks the quantum information, quantum gravity, and astrophysics communities for the foundational works on which this synthesis builds. Special thanks to the SPARC team for making their data publicly available.

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